HALTECH
MULTIPLEXED CAN PROTOCOL

DOCUMENT VERSION 2.0



THIS DOCUMENT DESCRIBES THE CAN PROTOCOL IMPLEMENTED WITHIN THE FOLLOWING HALTECH PRODUCTS:

Nexus PD16 Power Distribution Module
Nexus TI4L Transmission Interface
Nexus IO16 Input and Output Expander
Haltech iC-7 Display Dash
Haltech uC-10 Display Dash

OVERVIEW

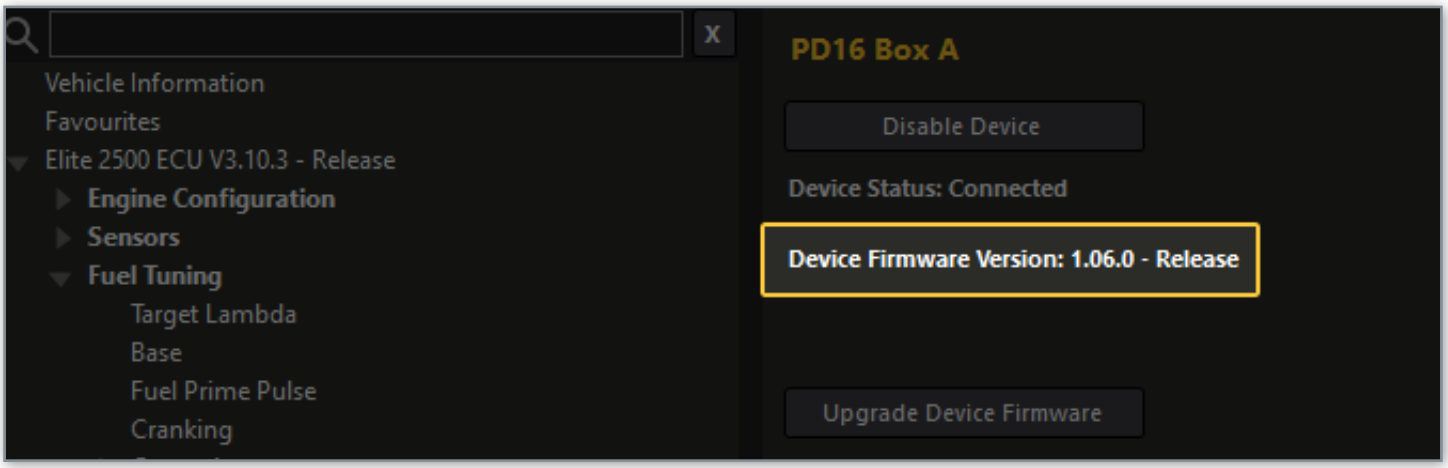
The Haltech Multiplexed CAN Protocol, previously known as the PD16 CAN protocol, enables seamless communication and sharing of inputs and outputs between Haltech Nexus and Elite ECUs and a variety of Haltech CAN devices. Originally designed for the PD16, the protocol now supports a broader range of devices, such as the Haltech uC-10 or iC-7 dash displays, transmission interfaces, and future IO expansion devices, allowing for real-time IO sharing over the CAN bus. This multiplexed approach simplifies wiring, reduces complexity, and enhances coordination between devices, making it ideal for both motorsport and performance vehicle applications.

DEVICE FIRMWARE

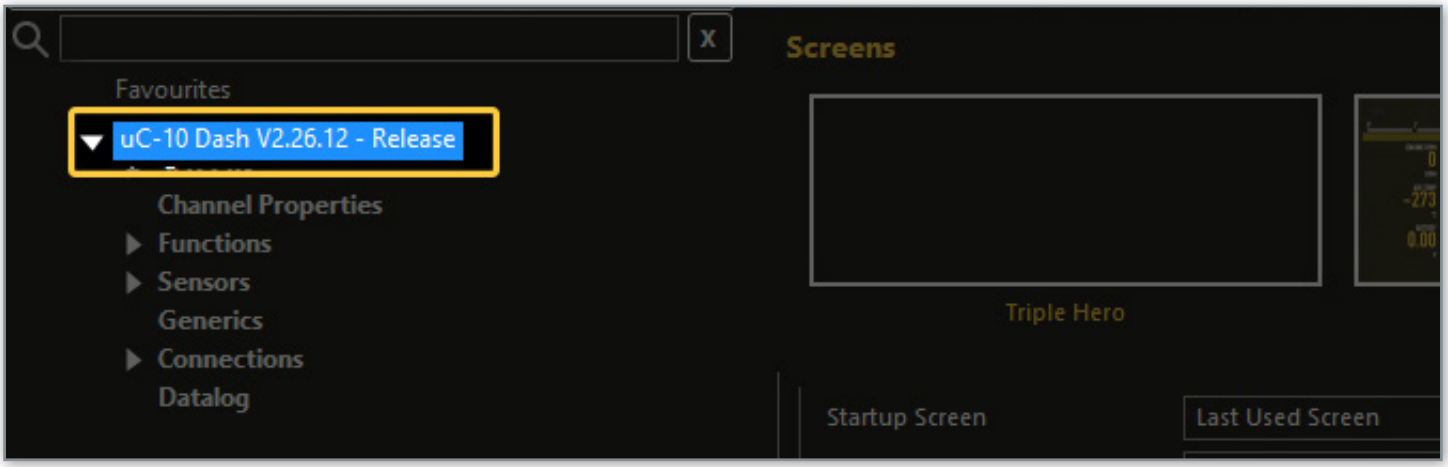
Please ensure that your Haltech CAN device is running the firmware version listed below or a later version.

Haltech CAN Device	Required Firmware Version
Nexus PD16	Version 1.09
Nexus TI4L	Version 1.10
Nexus IO16	Version 1.10
Haltech uC-10	Version 1.40
Haltech iC-7	Version 1.40

The PD16, TI4L or IO16 firmware version can be checked in the Nexus Software Programmer (NSP) while connected to your Haltech ECU.



The Haltech uC-10 or iC-7 display dash firmware version can be checked by connecting directly to the dash using the Nexus Software Programmer (NSP).



KEY INFORMATION

The Haltech CAN bus operates at 1 MBit and uses 11-bit IDs. IDs are expressed in Hexadecimal.

The first byte in a packet is considered byte 0, and the 8th byte is 7. The most significant bit in a byte is considered 7 and the least significant is bit 0.

Data is encoded as big-endian.

ADDRESSING DATA

Data that is the size of an individual byte (that is byte aligned) are represented with the byte number, e.g. 4 for data byte 4.

Data that crosses multiple bytes (that is byte aligned) are represented with a byte range. The range is inclusive, e.g. 0-1 for 2 bytes of data in bytes 0 and 1, 4-7 for 4 bytes of data in bytes 4, 5, 6 and 7.

To address data stored in individual bit within a byte, the following notation is used X:Y. The X is the byte number and Y is the bit number, e.g. 4:0 for bit 0 in byte 4.

To address data stored across many bits that may span bytes, the above notation is used, but expressed as a range. The range is inclusive, e.g. 6:3-7:0 for 12 bits of data starting at bit 3 on the byte 6 to bit 0 on byte 7.

Example:

Byte	0								1								2								3								4							
Bit	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0
Data	Manifold Pressure																Switch State	Re-served			Input Voltage								Status Enum											
Value	1013 (0x03F5)																1	0			3000 (0x0BB8)								200 (0xC8)											
Addressing	0 - 1																2:07	2:6-2:4			2:3-3:0								4											
	0	0	0	0	0	0	1	1	1	1	1	1	0	1	0	1	1	0	0	0	1	0	1	1	1	0	1	1	1	0	0	0	1	1	0	0	1	0	0	0

The “Conversion from Raw” column in the example protocol specification table below, shows how to convert the raw value in the message to a value known unit as indicated by the “Units” column.

The x symbol represents the raw value and y represents the value converted.

E.g. The “Throttle Position” channel has a raw value of 456. The conversion for it is y = x / 10. Therefore the converted value of y equals:

y = 456 / 10 = 45.6%

If other units are required, it is the responsibility of the device reading these values to perform these conversions. All pressures are absolute, and it is necessary to subtract 101.3 kPa from the final result if gauge pressure is desired.

CAN ID	Rate (Hz)	Direction	Message Position	Sign	Channel	Units	Conversion from Raw
0x360	50	TX	0 - 1	Unsigned	RPM	RPM	y = x
			2 - 3	Unsigned	Manifold Pressure	kPa (Abs)	y = x/10
			4- 5	Unsigned	Throttle Position	%	y = x/10
			6 - 7	Unsigned	Coolant Pressure	kPa	y = x/10 - 101.3
0x361	50	TX	0 - 1	Unsigned	Fuel Pressure	kPa	y = x/10 - 101.3
			2 - 3	Unsigned	Oil Pressure	kPa	y = x/10 - 101.3
			4 - 5	Unsigned	Engine Demand	%	y = x/10
			6 - 7	Unsigned	Wastegate Pressure	kPa	y = x/10 - 101.3

SUPPORTED DEVICES

Device	Base CAN ID	Firmware version required
Nexus PD16	PD16 A: 0x6D0	Version 1.09
	PD16 B: 0x6D8	
	PD16 C: 0x6E0	
	PD16 D: 0x6E8	
Nexus IO16	IO16 A: 0x6A8	Version 1.10
	IO16 B: 0x6B0	
Nexus TI4L	0x6B8	Version 1.10
Haltech Dash (iC-7, uC-10)	0x680	Version 1.40

NOTE: Not all devices support all of the functionality provided by the protocol. For example, some devices do not support CAN ID swapping. In these cases, explicit CAN IDs are provided per product to clearly identify which products are supported.

MULTIPLEXING

We're sending and receiving a lot of data. We'll multiplex our CAN messages to consume as few CAN IDs as possible. If a message is multiplexed then the first byte will be a multiplex ID used to identify the payload. The Multiplex ID has two sections - the IO type section and the IO index section.

For example, the CAN packet for turning on the 8A output 3 and 25A output 4 will have the same CAN ID and the same payload, but the multiplex ID will indicate which output the packet corresponds to.

ECU / Dash to Device

PD16 packets are only sent from the ECU or Dash if that pin is enabled and not in conflict.

High Priority / High Output Rate

NOTE: The ECU and dash can send messages asynchronously whenever there are changes of the I/O. So for example the ECU will send out the message for controlling HBO 1 (CAN ID=base CAN ID, Mux Type=HBO, Mux Index=0) at the base rate of 50ms. So if no changes on the pin occur, that is what can be measured for that particular combination of CAN ID, Mux Type, and Mux Index. If the function that is controlling that pin updates at a faster rate, then you can expect to see a higher rate of CAN messages to control HBO 1. The PD16 will do the same with its inputs, if nothing changes on them, they come out at the base rate. If they are changing more often than the frequency is increased.

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID)	20	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		0 = 25A
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = 25A-HCO1
							1 = 25A-HCO2
							2 = 25A-HCO3
							3 = 25A-HCO4
			1:7 - 2:0	Unsigned	25A <Mux ID> Duty Cycle	%	1000 = 100.0%
			3:7 - 4:0	Unsigned	25A <Mux ID> Frequency	Hz	1000 = 1000Hz
(Base CAN ID)	20	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		1 = 8A
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = 8A-HCO1
							1 = 8A-HCO2
							2 = 8A-HCO3
							3 = 8A-HCO4
							4 = 8A-HCO5
							5 = 8A-HCO6
							6 = 8A-HCO7
							7 = 8A-HCO8
							8 = 8A-HCO9
							9 = 8A-HCO10
			1:7 - 2:0	Unsigned	8A <Mux ID> Duty Cycle	%	1000 = 100.0%
			3:7 - 4:0	Unsigned	8A <Mux ID> Frequency	Hz	1000 = 1000Hz
(Base CAN ID)	20	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		2 = Half Bridge
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = HBO1
							1 = HBO2
			1:7 - 2:0	Unsigned	Half Bridge <Mux ID> Duty Cycle	%	1000 = 100.0%
			3:7 - 4:0	Unsigned	Half Bridge <Mux ID> Frequency	Hz	1000 = 1000Hz
			5:7 - 6:0	Unsigned	Half Bridge <Mux ID> Target Current	mA	1000 = 1000mA
(Base CAN ID)	20	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		6 = DPO
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = DPO1
							1 = DPO2
							...
			2:7 - 3:0	Unsigned	DPO <Mux ID> Duty Cycle	%	1000 = 100.0%
			4:7 - 5:0	Unsigned	DPO <Mux ID> Frequency	Hz	1000 = 1000Hz

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		0 = 25A
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = 25A-HCO1
							1 = 25A-HCO2
							2 = 25A-HCO3
							3 = 25A-HCO4
			1:7 - 1:5	Unsigned	25A <Mux ID> Retries		0 = never
							1 = 1
							2 = 2
							3 = 5
							4 = 10
							5 = 20
							6 = 30
							7 = always
			1:4 - 1:3	Unsigned	25A <Mux ID> Drive Type		0 = undefined
							1 = high side
							2 = both
							3 = open circuit
			1:2 - 1:1	Unsigned	25A <Mux ID> Safe State		0 = inactive
							1 = active
							2 = open circuit
			1:00	Unsigned	25A <Mux ID> Active State		0 = low
							1 = high
			2:7 - 2:0	Unsigned	25A <Mux ID> Retry Delay		1 = 1 second
							range: 0 - 240
			3:7 - 3:0	Unsigned	25A <Mux ID> Fuse Current		1 = 0.2A
							range = 0 - 250
			4:7 - 4:0	Unsigned	25A <Mux ID> Soft Start Current		1 = 0.2A
							range = 0 - 250
			5:7 - 5:0	Unsigned	25A <Mux ID> Soft Start Duration		1 = 0.1 seconds
							range = 0 - 250

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		1 = 8A
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = 8A-HCO1
							1 = 8A-HCO2
							2 = 8A-HCO3
							3 = 8A-HCO4
							4 = 8A-HCO5
							5 = 8A-HCO6
							6 = 8A-HCO7
							7 = 8A-HCO8
							8 = 8A-HCO9
							9 =8A-HCO10
			1:7 - 1:5	Unsigned	8A <Mux ID> Retries		0 = never
							1 = 1
							2 = 2
							3 = 5
							4 = 10
							5 = 20
							6 = 30
							7 = always
			1:4 - 1:3	Unsigned	8A <Mux ID> Drive Type		0 = undefined
							1 = undefined
							2 = high side
							3 = open circuit
			1:2 - 1:1	Unsigned	8A <Mux ID> Safe State		0 = inactive
							1 = active
							2 = open circuit
			1:00	Unsigned	8A <Mux ID> Active State		0 = low
							1 = high
			2:7 - 2:0	Unsigned	8A <Mux ID> Retry Delay		1 = 1 second
							range: 0 - 240
			3:7 - 3:0	Unsigned	8A <Mux ID> Fuse Current		1 = 0.2A
							range = 0 - 250

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		2 = Half Bridge
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = HBO1
							1 = HBO2
			1:7 - 1:5	Unsigned	Half Bridge <Mux ID> Retries		0 = never
							1 = 1
							2 = 2
							3 = 5
							4 = 10
							5 = 20
							6 = 30
							7 = always
			1:4 - 1:3	Unsigned	Half Bridge <Mux ID> Drive Type		0 = low side
							1 = high side
							2 = both
							3 = open circuit
			1:2 - 1:1	Unsigned	Half Bridge <Mux ID> Safe State		0 = inactive
							1 = active
							2 = open circuit
			1:00	Unsigned	Half Bridge <Mux ID> Active State		0 = low
							1 = high
			2:7 - 2:0	Unsigned	Half Bridge <Mux ID> Retry Delay		1 = 1 second
							range: 0 - 240
			3:00	Unsigned	Half Bridge <Mux ID> Control Method		0 = Duty Cycle
							1 = Current Control
			4:7 - 4:0	Unsigned	Half Bridge <Mux ID> Fuse Current		1 = 0.2A
							range = 0 - 250
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		3 = SPI
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = SPI1
							1 = SPI2
							2 = SPI3
							3 = SPI4
			1:7-1:6	Unsigned	SPI <Mux ID> Pin Mode		0 = Reluctor
							1 = Hall Effect
							2 = Custom
			1:5 - 1:4	Unsigned	SPI <Mux ID> Input State Startup State		0 = Off
							1 = On
							2 = Last State
							Note: Dash Only
			1:03	Unsigned	SPI <Mux ID> Input State Mode		0 = push on push off
							1 = momentary
							Note: Dash Only
			1:2 - 1:1	Unsigned	SPI <Mux ID> Pullup		0 = disable
							1 = enable
			1:00	Unsigned	SPI <Mux ID> Edge Select		0 = falling
							1 = rising
			4:7 - 5:0	Unsigned	SPI <Mux ID> Upper Threshold	mV	5000 = 5.00V
			6:7 - 7:0	Unsigned	SPI <Mux ID> Glitch Filter	us	1 = 1us
							Range: 0 - 500

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		4 = AVI
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = AVI1
							1 = AVI2
							2 = AVI3
							3 = AVI4
			1:5 - 1:4	Unsigned	AVI <Mux ID> Input State Startup State		0 = Off
							1 = On
							2 = Last State
							Note: Dash Only
			1:03	Unsigned	AVI <Mux ID> Input State Mode		0 = push on push off
							1 = momentary
							Note: Dash Only
			1:2 - 1:1	Unsigned	AVI <Mux ID> Pullup		0 = disable
							1 = enable (1k)
							2 = enable (240Ω) (AVI 1 only)
			2:7 - 3:0	Unsigned	AVI <Mux ID> Switch On Threshold	mV	5000 = 5.00V
			4:7 - 5:0	Unsigned	AVI <Mux ID> Switch Off Threshold	mV	5000 = 5.00V
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		5 = DPI
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = DPI1
							1 = DPI2
							...
			1:5 - 1:4	Unsigned	DPI <Mux ID> Input State Startup State		0 = Off
							1 = On
							2 = Last State
							Note: Dash Only
			1:03	Unsigned	DPI <Mux ID> Input State Mode		0 = push on push off
							1 = momentary
							Note: Dash Only
			1:1 - 1:2	Unsigned	DPI <Mux ID> Pullup		0 = disable
							1 = enable
			1:00	Unsigned	DPI <Mux ID> Edge Select		0 = falling
							1 = rising
			2:7 - 3:0	Unsigned	DPI <Mux ID> Upper Threshold	mV	5000 = 5.00V
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		6 = DPO
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = DPO1
							1 = DPO2
							...
			1:4 - 1:3	Unsigned	DPO <Mux ID> Pullup		0 = disable
							1 = 5V
							2 = Battery Voltage
			1:2 - 1:1	Unsigned	DPO <Mux ID> Safe State		0 = inactive
							1 = active
							2 = open circuit
			1:00	Unsigned	DPO <Mux ID> Active State		0 = low
							1 = high

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 1	2	RX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		7 = Button
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = Button 0 - 6 1 = Button 7 - 12 ...
			1:5 - 1:4	Unsigned	Button 1 + <Mux ID> Input State Startup State		0 = Off 1 = On 2 = Last State Note: Dash Only
			1:03	Unsigned	Button 1 + <Mux ID> Input State Mode		0 = push on push off 1 = momentary Note: Dash Only
			2:5 - 2:4	Unsigned	Button 2 + <Mux ID> Input State Startup State		0 = Off 1 = On 2 = Last State Note: Dash Only
			2:03	Unsigned	Button 2 + <Mux ID> Input State Mode		0 = push on push off 1 = momentary Note: Dash Only
			3:5 - 3:4	Unsigned	Button 3 + <Mux ID> Input State Startup State		0 = Off 1 = On 2 = Last State Note: Dash Only
			3:03	Unsigned	Button 3 + <Mux ID> Input State Mode		0 = push on push off 1 = momentary Note: Dash Only
			4:5 - 4:4	Unsigned	Button 4 + <Mux ID> Input State Startup State		0 = Off 1 = On 2 = Last State Note: Dash Only
			4:03	Unsigned	Button 4 + <Mux ID> Input State Mode		0 = push on push off 1 = momentary Note: Dash Only
			5:5 - 5:4	Unsigned	Button 5 + <Mux ID> Input State Startup State		0 = Off 1 = On 2 = Last State Note: Dash Only
			5:03	Unsigned	Button 5 + <Mux ID> Input State Mode		0 = push on push off 1 = momentary Note: Dash Only
			6:5 - 6:4	Unsigned	Button 6 + <Mux ID> Input State Startup State		0 = Off 1 = On 2 = Last State Note: Dash Only
			6:03	Unsigned	Button 6 + <Mux ID> Input State Mode		0 = push on push off 1 = momentary Note: Dash Only

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 2	2	RX	0:04	Unsigned	ECR status*		0 = off
							1 = on
							*ECR, when on, will force the PD16 to stay on even if the ignition power switch is removed (until either this bit gets sent with a zero, or CAN comms is lost). Only when in CAN master mode (ie, device was turned on with the ignition switch).
			0:03	Unsigned	CAN Power		0 = disabled
							1 = enabled*
							*Not Implemented / Reserve
			0:02	Unsigned	Reserved		0 = disabled
			0:01	Unsigned	CAN Termination (Bus 1)		0 = disabled
							1 = enabled
			0:00	Unsigned	Reserved		0 = disabled

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
PD16 A: 0x320 PD16 B: 0x321 PD16 C: 0x322 PD16 D: 0x323 IO16 A: 0x330 IO16 B: 0x334 TI4L: 0x338 Dash: 0x340	100	TX	0:07	Unsigned	AVI 1 State		0 = OFF
							1 = ON
			0:6 - 1:0	Unsigned	AVI 1 Voltage	mV	5000 = 5.00V
			2:07	Unsigned	AVI 2 State		0 = OFF
							1 = ON
			2:6 - 3:0	Unsigned	AVI 2 Voltage	mV	5000 = 5.00V
			4:07	Unsigned	AVI 3 State		0 = OFF
							1 = ON
			4:6 - 5:0	Unsigned	AVI 3 Voltage	mV	5000 = 5.00V
			6:07	Unsigned	AVI 4 State		0 = OFF
							1 = ON
			6:6 - 7:0	Unsigned	AVI 4 Voltage	mV	5000 = 5.00V
IO16 A: 0x331 IO16 B: 0x335 TI4L: 0x339 Dash: 0x341	100	TX	0:07	Unsigned	AVI 5 State		0 = OFF
							1 = ON
			0:6 - 1:0	Unsigned	AVI 5 Voltage	mV	5000 = 5.00V
			2:07	Unsigned	AVI 6 State		0 = OFF
							1 = ON
			2:6 - 3:0	Unsigned	AVI 6 Voltage	mV	5000 = 5.00V
			4:07	Unsigned	AVI 7 State		0 = OFF
							1 = ON
							Dash Only
			4:6 - 5:0	Unsigned	AVI 7 Voltage	mV	5000 = 5.00V
							Dash Only
			6:07	Unsigned	AVI 8 State		0 = OFF
							1 = ON
							Dash Only
			6:6 - 7:0	Unsigned	AVI 8 Voltage	mV	5000 = 5.00V
							Dash Only
Dash: 0x342	100	TX	0:07	Unsigned	AVI 9 State		0 = OFF
							1 = ON
			0:6 - 1:0	Unsigned	AVI 9 Voltage	mV	5000 = 5.00V
			2:07	Unsigned	AVI 10 State		0 = OFF
							1 = ON
			2:6 - 3:0	Unsigned	AVI 10 Voltage	mV	5000 = 5.00V
PD16 A: 0x324 PD16 B: 0x325 PD16 C: 0x326 PD16 D: 0x327 IO16 A: 0x332 IO16 B: 0x336 TI4L: 0x33A Dash: 0x343	100	TX	0:07	Unsigned	SPI 1 State		0 = OFF
							1 = ON
			0:6 - 1:0	Unsigned	SPI 1 Voltage	mV	5000 = 5.00V
							No IC-7 Dash support
			2:07	Unsigned	SPI 2 State		0 = OFF
							1 = ON
			2:6 - 3:0	Unsigned	SPI 2 Voltage	mV	5000 = 5.00V
							No IC-7 Dash support
			4:07	Unsigned	SPI 3 State		0 = OFF
							1 = ON
			4:6 - 5:0	Unsigned	SPI 3 Voltage	mV	5000 = 5.00V
							No IC-7 Dash support
			6:07	Unsigned	SPI 4 State		0 = OFF
							1 = ON
			6:6 - 7:0	Unsigned	SPI 4 Voltage	mV	5000 = 5.00V
							No IC-7 Dash support

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
PD16 A: 0x328 PD16 B: 0x329 PD16 C: 0x32A PD16 D: 0x32B IO16 A: 0x333 IO16 B: 0x337 TI4L: 0x33B Dash: 0x344	100	TX	0:7 - 1:0	Unsigned	SPI 1 Frequency	Hz	1 = 0.25Hz
			2:7 - 3:0	Unsigned	SPI 2 Frequency	Hz	1 = 0.25Hz
			4:7 - 5:0	Unsigned	SPI 3 Frequency	Hz	1 = 0.25Hz
			6:7 - 7:0	Unsigned	SPI 4 Frequency	Hz	1 = 0.25Hz

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 3	20	TX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		3 = SPI
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = SPI1
							1 = SPI2
							2 = SPI3
							3 = SPI4
			1:00	Unsigned	SPI <Mux ID> State		0 = OFF
							1 = ON
			2:7 - 3:0	Unsigned	SPI <Mux ID> Voltage	mV	5000 = 5.000V
			4:7 - 5:0	Unsigned	SPI <Mux ID> Duty Cycle	%	1000 = 100.0%
			6:7 - 7:0	Unsigned	SPI <Mux ID> Frequency	Hz	1000 = 1000Hz
(Base CAN ID) + 3	20	TX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		4 = AVI
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = AVI1
							1 = AVI2
							2 = AVI3
							3 = AVI4
			1:00	Unsigned	AVI <Mux ID> State		0 = OFF
							1 = ON
			2:7 - 3:0	Unsigned	AVI <Mux ID> Voltage	mV	5000 = 5.000V
(Base CAN ID) + 3	20	TX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		5 = DPI
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = DPI1
							1 = DPI2
							...
			1:00	Unsigned	DPI <Mux ID> State		0 = OFF
							1 = ON
			4:7 - 5:0	Unsigned	DPI <Mux ID> Duty Cycle	%	1000 = 100.0%
			6:7 - 7:0	Unsigned	DPI <Mux ID> Frequency	Hz	1000 = 1000Hz
(Base CAN ID) + 3	20	TX	0:7 - 0:5	Unsigned	Mux ID (IO Index)		7 = Button
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = Button 0-6
			1:7 - 1:0	Signed	Button 0 state		0 = OFF
							1 = ON
							< 0 is Pinerror
			2:7 - 2:0	Signed	Button 1 state		0 = OFF
							1 = ON
							< 0 is Pinerror
			3:7 - 3:0	Signed	Button 2 state		0 = OFF
							1 = ON
							< 0 is Pinerror
			4:7 - 4:0	Signed	Button 3 state		0 = OFF
							1 = ON
							< 0 is Pinerror
			5:7 - 5:0	Signed	Button 4 state		0 = OFF
							1 = ON
							< 0 is Pinerror
			6:7 - 6:0	Signed	Button 5 state		0 = OFF
							1 = ON
							< 0 is Pinerror

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 4	5	TX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		0 = 25A
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = 25A-HCO1
							1 = 25A-HCO2
							2 = 25A-HCO3
							3 = 25A-HCO4
			1:7 - 1:0	Unsigned	25A <Mux ID> Load	%	100 = 100%
			2:7 - 3:0	Unsigned	25A <Mux ID> Voltage	mV	12000 = 12.000V
			4:7 - 5:0	Unsigned	25A <Mux ID> Low Side Current	mA	50000 = 50.000A
			7:7 - 6:0	Unsigned	25A <Mux ID> High Side Current	mA	50000 = 50.000A
(Base CAN ID) + 4	5	TX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		1 = 8A
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = 8A-HCO1
							1 = 8A-HCO2
							2 = 8A-HCO3
							3 = 8A-HCO4
							4 = 8A-HCO5
							5 = 8A-HCO6
							6 = 8A-HCO7
							7 = 8A-HCO8
							8 = 8A-HCO9
							9 =8A-HCO10
			1:7 - 1:3	Unsigned	8A <Mux ID> Retry Count		0 - 31
							31 indicates > 30
			1:2 - 1:0	Unsigned	8A <Mux ID> Pin State		0 = operational
							1 = high current
							2 = over current
							3 = short circuit
							4 = open circuit
							5 = Invalid Pin Selection
							6 = Unexpected Profile
							7 = Disabled
			2:7 - 3:0	Unsigned	8A <Mux ID> Voltage	mV	12000 = 12.000V
			4:7 - 5:0	Unsigned	8A <Mux ID> Current	mA	1000 = 1.000A
			6:7 - 6:0	Unsigned	8A <Mux ID> Load	%	100 = 100%
			7:7 - 7:3	Unsigned	25A <Mux ID> Retry Count	(for HCO8 1-4 only)	0 - 31
							31 indicates > 30
			7:2 - 7:0	Unsigned	25A <Mux ID> Pin State	(for HCO8 1-4 only)	0 = operational
							1 = high current
							2 = over current
							3 = short circuit
							4 = open circuit
							5 = Invalid Pin Selection
							6 = Unexpected Profile
							7 = Disabled

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 4	5	TX	0:7 - 0:5	Unsigned	Mux ID (IO Type)		2 = Half Bridge
			0:3 - 0:0	Unsigned	Mux ID (IO Index)		0 = HBO1
							1 = HBO2
			1:7 - 1:3	Unsigned	Half Bridge <Mux ID> Retry Count		0 - 31
							31 indicates > 30
			1:2 - 1:0	Unsigned	Half Bridge <Mux ID> Pin State		0 = operational
							1 = high current
							2 = over current
							3 = short circuit
							4 = open circuit
							5 = Invalid Pin Selection
							6 = Unexpected Profile
							7 = Disabled
			2:7 - 3:0	Unsigned	Half Bridge Voltage	mV	12000 = 12.000V
			4:7 - 4:0	Unsigned	Half Bridge <Mux ID> Load	%	100 = 100%
			5:7 - 5:0	Reserved	Reserved	Reserved	Reserved
			6:7 - 7:0	Unsigned	Half Bridge HS Current		
(Base CAN ID) + 5	2	TX	0:7 - 0:4	Unsigned	Status		0=In bootmode
							1=In firmware
							2=Hardware failure
							3=Firmware erased
							4=Watchdog timeout
							5=Illegal op-code
			0:03	Unsigned	USB Connected		
			0:01	Unsigned	ID Conflict		
			1:7 - 1:3	Unsigned	Boot Version		
			1:1 - 1:0	Unsigned	FW Major Version		
			2:7 - 2:0	Unsigned	FW Minor Version		
			3:7 - 3:0	Unsigned	FW Bugfix Version		
			4:7 - 4:0	Unsigned	FW Release Version		
			5:7 - 5:0	Unsigned	Device Variant		Dash Only
(Base CAN ID) + 6	2	TX	0:7 - 0:0	Unsigned	Mux ID = 0		
			1:00	Unsigned	PD16: Ignition input switch status		PD16
					IO16: Const power input status		0 = off
							1 = on
							IO16
							0 = CAN power
							1 = const power
			2:7 - 3:0	Unsigned	Main Rail Voltage	mV	12000 = 12.000V
			4:7 - 5:0	Unsigned	Protected Rail Voltage	mV	12000 = 12.000V
			6:7 - 7:0	Unsigned	ECR Plug Voltage	mV	12000 = 12.000V
							PD16 Only

CAN I.D.	Rate (Hz)	Direction	Message position	Sign	Channel	Units	Conversion from raw
(Base CAN ID) + 6	2	TX	0:7 - 0:0	Unsigned	Mux ID = 1		
			2:7 - 3:0	Unsigned	Boost Voltage	mV	12000 = 12.000V
							The expected value should sit 10V above the Battery Voltage
			4:7 - 5:0	Unsigned	SEPIC Voltage	mV	12000 = 12.000V
							An internal rail. The expected value is 12V
			6:7 - 7:0	Unsigned	ECR Plug Current	mA	1 = 1mA
							PD16 Only
(Base CAN ID) + 6	2	TX	0:7 - 0:0	Unsigned	Mux ID = 2		
			2:7 - 3:0	Unsigned	UVLO Voltage	mV	12000 = 12.000V
							An internal rail. The expected value is 3.3V
			4:7 - 5:0	Unsigned	Sensor Ground Voltage	mV	12000 = 12.000V
			6:7 - 7:0	Unsigned	VDD Voltage	mV	12000 = 12.000V
(Base CAN ID) + 6	2	TX	0:7 - 0:0	Unsigned	Mux ID = 3		
			1:7 - 1:0	Unsigned	Mail Rail Temperature		1 = 1 degree Celsius
			2:7 - 2:0	Unsigned	Thermistor 1 Temperature		Temperatures below zero read as zero.
			3:7 - 3:0	Unsigned	Thermistor 2 Temperature		PD16 Only
			4:7 - 4:0	Unsigned	Thermistor 3 Temperature		
			5:7- 5:0	Unsigned	CPU Temperature		
			6:7 - 7:0	Unsigned	Total Current Draw	A	0.1 A resolution
							1000 = 100.0A
(Base CAN ID) + 6	2	TX	0:7 - 0:0	Unsigned	Mux ID = 4		
			2:7 - 3:0	Unsigned	Battery Voltage	mV	12000 = 12.000V
			4:7 - 5:0	Unsigned	On Time		1 = 1 second
			6:7 - 6:6	Unsigned	HCO25 3 & 4 Temperature		0 = OK
			6:5 - 6:4	Unsigned	HCO25 1 & 2 Temperature		1 = Warning (> 115 degrees)
			6:3 - 6:2	Unsigned	TVS Temperature		2 = shutdown (> 145 degrees)
			6:1 - 6:0	Unsigned	Main Rail Temperature		PD16 Only
(Base CAN ID) + 6	2	TX	0:7 - 0:0	Unsigned	Mux ID = 5		
			2:7 - 2:4	Unsigned	Digit 1 of serial number		0x19
			2:3 - 2:0	Unsigned	Digit 2 of serial number		
			3:7 - 3:4	Unsigned	Digit 3 of serial number		0x80
			3:3 - 3:0	Unsigned	Digit 4 of serial number		
			4:7 - 4:4	Unsigned	Digit 5 of serial number		0x00
			4:3 - 4:0	Unsigned	Digit 6 of serial number		
			5:7 - 5:4	Unsigned	Digit 7 of serial number		0x1c
			5:3 - 5:0	Unsigned	Digit 8 of serial number		
			6:7 - 6:4	Unsigned	Digit 9 of serial number		0x3c
			6:3 - 6:0	Unsigned	Digit 10 of serial number		
			7:7 - 7:4	Unsigned	Digit 11 of serial number		0xd1
			7:3 - 7:0	Unsigned	Digit 12 of serial number		

REQUEST TO SWAP ID (FROM ECU):

CAN ID	Byte pos	Value
PD16 A: 0x6F7 PD16 B: 0x6F9 PD16 C: 0x6Fb PD16 D: 0x6Fb IO16 A: 0x6CA IO16 B: 0x6CC	0	0x04
	1	0x2e
	2	0x43
	3	0x2b
	4	Value of the new box ID from 0 - 3
	5	0x55
	6	0x55
	7	0x55

PD16 RESPONSE / ACKNOWLEDGEMENT:

CAN ID	Byte pos	Value
PD16 A: 0x6F8 PD16 B: 0x6FA PD16 C: 0x6FC PD16 D: 0x6FE IO16 A: 0x6CB IO16 B: 0x6CD	0	0x01
	1	0x6e
	2	0xaa
	3	0xaa
	4	0xaa
	5	0xaa
	6	0xaa
	7	0xaa

REQUEST SOFT RESET:

CAN ID	Byte pos	Value
PD16 A: 0x6F7 PD16 B: 0x6F9 PD16 C: 0x6FB PD16 D: 0x6FD IO16 A: 0x6CA IO16 B: 0x6CC	0	0x02
	1	0x11
	2	0x03 (soft reset) (could be 01 for hard reset or 02 for key on/off reset)
	3	0x55
	4	0x55
	5	0x55
	6	0x55
	7	0x55

PD16 RESET RESPONSE / ACKNOWLEDGEMENT:

CAN ID	Byte pos	Value
PD16 A: 0x6F8 PD16 B: 0x6FA PD16 C: 0x6FC PD16 D: 0x6FE IO16 A: 0x6CB IO16 B: 0x6CD	0	0x03
	1	0x51
	2	0x03
	3	0x00
	4	0xaa
	5	0xaa
	6	0xaa
	7	0xaa

CAN ID	Payload	Number of messages	Desired output rate (ms)	Cycle time (ms)	CAN packets/s
Base CAN ID	• Output duty cycle	16	50	3	333
	• Output frequency				
Base CAN ID + 1	• Output settings	24	500	20	50
	• Input settings				
Base CAN ID + 2	• System config	1	500	500	2
Base CAN ID + 3	• Input state	8	50	6.25	160
	• Input voltage				
	• Input duty cycle				
	• Input frequency				
Base CAN ID + 4	• Output voltage	16	200	12.5	80
	• Output retry count				
	• Output pin state				
Base CAN ID + 5	• Status info	1	500	500	2
Base CAN ID + 6	• Serial number	1	500	500	2
	• Battery voltage				
	• Diagnostics				
					629

Thank you for your support of Haltech Engine Management and our CAN based devices.

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